ABU NASIR PATWARY

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EDUCATION

United International University **Nov 2021 – Nov 2026**

B.Sc. in Computer Science and Engineering | CGPA: 3.22/4.00

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Java

Familiar With: JavaScript, PHP

Frameworks: Django (currently learning)

Databases: MySQL, PostgreSQL

Data Analysis & Visualization: NumPy, Pandas, Matplotlib, Seaborn, Plotly

Developer Tools: Git, Linux (basic familiarity)

PROJECTS

Khabo-Koi — Restaurant Discovery & Table Reservation Platform

Technologies: React.js, Vite, Tailwind CSS, Django REST Framework, PostgreSQL

- A full-stack restaurant discovery and reservation platform for exploring restaurants, viewing menus, checking table availability, and managing bookings.
- Integrates food preordering and an AI dining assistant, with role-based dashboards for customers, restaurant managers, and administrators.

CampusLink — University Community Platform

Technologies: React.js, Vite, Tailwind CSS, Spring Boot, REST APIs, PostgreSQL

- A centralized university community platform designed to connect students, alumni, and administrators through integrated campus services.
- Brings together lost-and-found management, borrowing requests, campus complaints, and talent sharing, with notifications, role-based access, and content moderation.

RESEARCH PAPERS & PUBLICATIONS

Workload-Based Energy Efficiency Analysis of Native and Virtualized Operating Systems for Sustainable Computing: An Experimental Study on Modern Computing Platforms

Submitted — IEEE Computer Society Bangladesh Chapter Symposium 2026

- Evaluated energy efficiency across native macOS and virtualized Windows and Ubuntu Linux on Apple M3 hardware using four computational workloads and EnergiBridge measurements.
- Analyzed power consumption, energy usage, statistical variability, and estimated carbon footprint, identifying workload-dependent differences in operating system efficiency.

RESEARCH PAPERS & PUBLICATIONS (CONTINUED)

Colorism in Vision-Language Models: A Bangladeshi Study of Skin-Tone and Gender Bias in VLMs

Undergraduate Thesis | Manuscript in Preparation

- Conducted a Bangladesh-focused fairness evaluation of four vision-language models using 100 synthetic facial images representing 50 intended light- and dark-skin identity pairs.
- Analyzed 1,200 generated socioeconomic profiles and 4,000 simulated recruitment decisions, identifying model-specific skin-tone-associated disparities and candidate-position effects.

Robustness and Enhancement Recovery of CNN and Vision Transformer Models for Rice Leaf Disease Classification Under Controlled Image Degradation

Manuscript in Preparation

- Compared MobileNetV3-Small and DeiT-Tiny for seven-class rice leaf disease classification under controlled image noise, blur, and low-light conditions.
- Evaluated image enhancement techniques using accuracy, macro-F1, PSNR, and SSIM, demonstrating that improved image quality does not always translate into better classification performance.

Empirical Evaluation of Living-off-the-Land (LotL) Attack Techniques and Security Control Limitations in Modern Operating Systems

Manuscript in Preparation

- Investigated contextual detection of Living-off-the-Land techniques across Windows and macOS using a catalog of 40 techniques and 200 synthetically constructed evaluation traces.
- Developed and evaluated two rule-based detection approaches using native-tool activity and contextual features, including execution sequences, timing, process relationships, network activity, and persistence behavior, across Windows and macOS.

HONORS & AWARDS

Academic Merit Scholarship — Spring 2026

United International University, Dhaka, Bangladesh

- Awarded in recognition of academic performance during the Spring 2026 semester.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Joint Secretary (Administration) — 13th Executive Committee

UIU Debate Club, United International University

- Served as Joint Secretary (Administration), contributing to the club's administrative and organizational activities.
- Participated in two national-level debate competitions.